HUSSAIN HABIB

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EDUCATION

Lahore University of Management Sciences (LUMS) *2023 - May 2027*
B.S. in Computer Science, Minor in Robotics **CGPA: 3.79/4.00**
Dean's Honor List every year (Freshman, Sophomore, Junior)

Relevant Coursework: Computer Vision, Machine Learning, Deep Learning, Mobile Robotics, Microcontrollers & Interfacing, Multi-Agent Systems, Junior Design Studio (Robotics)

PUBLICATIONS

UAV-Based Crop Reconstruction and Trait Estimation Using RGB Imagery Without External Geospatial Infrastructure
Muhammad Hussain Habib Chaudhry, Muhammad Ibrahim Rana, and Hassan Jaleel
Computers and Electronics in Agriculture, Elsevier, 2026. DOI: 10.1016/j.compag.2026.111677

MANUSCRIPTS UNDER REVIEW

Hive-Nav: Belief-Guided Multi-Robot Room Classification and Open-Vocabulary Object Search
Muhammad Hussain Habib Chaudhry*, Umar Habib*, and Hassan Jaleel (*equal contribution)
Submitted to IEEE International Conference on Robotics and Automation (ICRA), 2027

FARMS: Field-Aligned RGB-LiDAR Monitoring System for Longitudinal UAV Crop Phenotyping
Co-author. Submitted to IEEE International Conference on Robotics and Automation (ICRA), 2027

RESEARCH EXPERIENCE

Lahore University of Management Sciences June 2024 - Present
Research Assistant, Robotics @ NARL *Supervisor: Dr. Hassan Jaleel*

- **Hive-Nav:** Led Gazebo simulation, designed cross-robot belief fusion and a disagreement-triggered revisit protocol, and handled networking for three RoboMaster EP robots (Pi 5, ROS 2, Nav2) on one GPU. Replaced unreliable VLM room labels with object inventories scored by a signed evidence table, raising real-world room accuracy from 88.3% (single visit) to 98.3% (three robots). Benchmarked 11 VLMs (Qwen3-VL-4B: 94.5% at 3.3 GB VRAM); after measuring up to 20.8% Wi-Fi packet loss, replaced streaming with on-demand frames with no new hardware.
- **UAV phenotyping:** Flew weekly DJI Mini 4 Pro surveys of two rice plots for a season and built the Metashape photogrammetry workflow (orthomosaics, DEMs) without RTK-GPS or GCPs. Traced a 1.09 m DEM doming artifact to near-parallel viewing geometry and wide-angle distortion, and designed a yawing multi-S flight plan that halved it (0.53 m); with RANSAC plane correction, canopy heights matched manual measurements (R = 0.876, MAE 0.177 m). Plant counting reached 94.95% accuracy
- **FARMS:** Built the RGB pipeline for a 16-plot wheat field, from orthomosaics to automatic plot recovery (median IoU 0.867 vs. manual plots), stress-tested with threshold ablations. RGB-LiDAR fusion for variety clustering: matched or beat RGB alone on all 7 dates, 100% on the last 2 dates

TEACHING EXPERIENCE

Microcontrollers and Interfacing (EE-324) August 2026 - Present
Teaching Assistant (50 students) *Instructor: Dr. Hafsa Qamar*

- Design and grade quizzes and homework assignments, and hold 3 hours of office hours per week
- Designed a designette joining students from three courses: a 100 Hz Butterworth low-pass filter, Tustin-discretized at 10 kHz on a 12-bit ADC, built on paper, in Simulink and on hardware

Computational Problem Solving (CS-100) Summers 2025 & 2026
Teaching Assistant (70+ students): labs, office hours, quizzes and grading *Instructor: Dr. Malik Jahan*

SELECTED PROJECTS

AgriJEPA: Self-Supervised Latent World Models for Plant Growth July 2026 - Present
Senior Year Project *Supervisor: Dr. Hassan Jaleel*

- Built a JEPA world model forecasting plant growth in latent space from multi-view images, with a trait-anchored subspace; four views cut prediction error 39% over one, and rollouts hold $R^2 > 0.93$ to 10 days
- Six crop datasets, pre-registered gates, plant-clustered bootstrap CIs; calendar-clock baselines showed most apparent field-data gains were cohort identification, not growth

Localization Under Change: Repairing Stale Maps with Pose Graphs Fall 2026
EE565/CS593 Mobile Robotics *Supervisor: Dr. Talha Manzoor*

- Repairing stale office maps without ground truth (LILocBench): an SE(2) pose graph over odometry, ICP, loop closures and localizer poses, solved by a Gauss-Newton optimizer written from scratch
- Global localization 86.7% on the repaired map vs. 65.6% on the stale map

3D Scene Reconstruction and Virtual Tour Fall 2025
CS436 Computer Vision

- Structure-from-Motion from scratch (SIFT, PnP, triangulation, bundle adjustment): ~200k points

OPEN-SOURCE SOFTWARE

BAGEL: Browser-Native ROS Bag Visualizer 2026 - Present
github.com/Hussain004/BAGEL · bagel-ros2.vercel.app

- Designed a client-side viewer for ROS 1/2 bags (`.mcap`, `.db3`, `.bag`), PX4 logs, point clouds and Gaussian splats, with URDF overlays and live streaming; no install or upload needed

SAMOSA: Step Auditor for Matrix Operations, Stochastics & Algebra 2026 - Present
github.com/Hussain004/SAMOSA · samosa-math.vercel.app

- Directed a checker that verifies each LaTeX derivation step exactly over finite fields, with counterexamples; web app, CLI, GitHub Action, MCP server; 1,400+ npm downloads till date

Forma: Browser-Native ONNX/TFLite Model Visualizer & Editor 2026 - Present
github.com/Hussain004/Forma: graph editing, quantization, benchmarks

TECHNICAL SKILLS

Programming	Python, C++, Java, TypeScript, MATLAB, React, Three.js, Git
Robotics	ROS 2, Nav2, SLAM Toolbox, Gazebo, Simulink, Raspberry Pi, SolidWorks
Perception & 3D	PyTorch, OpenCV, YOLO-World/YOLO-E, SAM, VLMs, Agisoft Metashape